

System of Lake Pollution Mathematical Modeling & Analytical Solutions

Coupled ODE Systems • Compartmental Analysis • Matrix
Exponential • Clean-Up Dynamics

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1. Physical Description & Compartmental Principle

The **System of Lake Pollution** is a classical application of first-order coupled linear ordinary differential equations. It models how contaminants (such as industrial effluent, pesticides, or heavy metals) disperse, mix, and wash out through a network of interconnected lakes, channels, and reservoirs (notably inspired by the Laurentian Great Lakes system).

Definition — Well-Mixed Lake Compartment

Let V_i denote the constant volume of Lake i (in m^3 or km^3), and let $x_i(t)$ represent the total mass of pollutant present in Lake i at time t (in kg or metric tons). The uniform concentration of pollutant $c_i(t)$ throughout Lake i is:

$$c_i(t) = \frac{x_i(t)}{V_i} \quad [\text{mass/volume}]. \quad (1)$$

The well-mixed hypothesis assumes that the pollutant disperses instantaneously and homogeneously throughout the entire lake volume.

Mathematical Law / Theorem — Conservation of Pollutant Mass

For any lake compartment i , the time rate of change of total pollutant mass equals the rate at which pollutant enters minus the rate at which pollutant leaves:

$$\frac{dx_i}{dt} = \text{Rate}_{\text{in}} - \text{Rate}_{\text{out}}. \quad (2)$$

If water flows into Lake i from Lake j at volumetric flow rate F_{ji} with concentration $c_j(t)$, and clean/polluted water enters from an external source at rate $F_{\text{ext},i}$ with concentration $c_{\text{ext},i}(t)$, then:

$$\text{Rate}_{\text{in}} = \sum_{j \neq i} F_{ji} \frac{x_j(t)}{V_j} + F_{\text{ext},i} c_{\text{ext},i}(t). \quad (3)$$

Similarly, if water flows out of Lake i at total outflow rate $F_{\text{out},i}$, the mass outflow rate is:

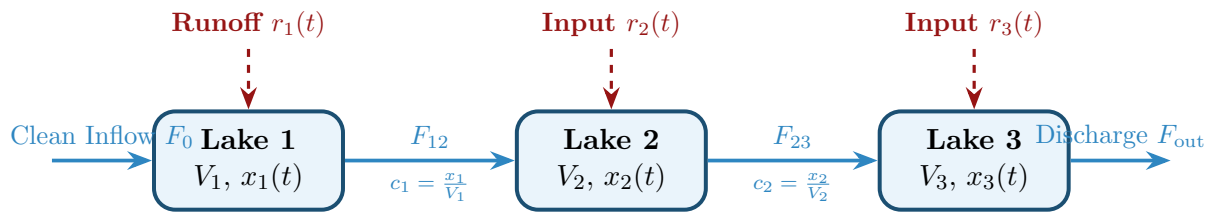
$$\text{Rate}_{\text{out}} = F_{\text{out},i} c_i(t) = \frac{F_{\text{out},i}}{V_i} x_i(t). \quad (4)$$

Hydrodynamic Equilibrium (Constant Volume):

To maintain constant lake volumes, the total volumetric water inflow must equal the total volumetric outflow for each lake:

$$\sum_{\text{in}} F = \sum_{\text{out}} F \implies \frac{dV_i}{dt} = 0.$$

The ratio $k_i = \frac{F_{\text{out},i}}{V_i}$ has dimensions of [1/time] and represents the **turnover rate** (reciprocal of the mean residence time $\tau_i = V_i/F_{\text{out},i}$).



2. Matrix State-Space Representation

Derivation & Formulation — General System Formulation for n Lakes

For a network of n interconnected lakes, the mass of pollutant in each lake is governed by the non-homogeneous linear first-order system:

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}(t) + \mathbf{r}(t), \quad \mathbf{x}(0) = \mathbf{x}_0, \quad (5)$$

where:

- $\mathbf{x}(t) = [x_1(t) \ x_2(t) \ \cdots \ x_n(t)]^T$ is the state vector of pollutant masses.
- $\mathbf{r}(t) = [r_1(t) \ r_2(t) \ \cdots \ r_n(t)]^T$ represents external contaminant influx rates (mass/time).
- $A \in \mathbb{R}^{n \times n}$ is the **transfer matrix** defined by:

$$A_{ii} = -\frac{F_{\text{out},i}}{V_i}, \quad A_{ij} = \frac{F_{ji}}{V_j} \quad (i \neq j). \quad (6)$$

Key Governing Equation**Analytical Solution via Matrix Exponential:**

For constant external input vector $\mathbf{r}$ and invertible transfer matrix A :

$$\mathbf{x}(t) = e^{At} \mathbf{x}_0 + A^{-1} (e^{At} - I) \mathbf{r} = e^{At} (\mathbf{x}_0 - \mathbf{x}^*) + \mathbf{x}^*, \quad (7)$$

where $\mathbf{x}^* = -A^{-1}\mathbf{r}$ is the **steady-state equilibrium distribution**.

Asymptotic Stability:

Because each diagonal entry $A_{ii} = -\frac{F_{\text{out},i}}{V_i} < 0$ and the sum of column entries satisfies $\sum_i A_{ij} \leq 0$, by the Gershgorin Disc Theorem all eigenvalues of A have strictly negative real parts:

$$\text{Re}(\lambda_k) < 0 \quad \forall k \in \{1, 2, \dots, n\}.$$

Consequently, $e^{At} \rightarrow 0$ as $t \rightarrow \infty$, proving that the system is **unconditionally asymptotically stable**. Any transient perturbation dies out exponentially.

3. Case I: Two Interconnected Lakes with Bidirectional Flow

Consider two adjacent lakes with constant volumes V_1 and V_2 . Water circulates between them:

- Clean fresh water enters Lake 1 at rate F_0 .
- Water flows from Lake 1 to Lake 2 at rate F_{12} .
- A feedback flow returns water from Lake 2 to Lake 1 at rate F_{21} .
- Outflow leaves Lake 2 into a river at rate $F_e = F_0$.

Derivation & Formulation — Flow Balance & System Equations

Hydrodynamic volume conservation dictates:

$$\text{Lake 1: } F_0 + F_{21} = F_{12}, \quad \text{Lake 2: } F_{12} = F_{21} + F_e.$$

The rate of change of pollutant mass in each lake is:

$$\frac{dx_1}{dt} = r_1 - \frac{F_{12}}{V_1}x_1 + \frac{F_{21}}{V_2}x_2, \quad (8)$$

$$\frac{dx_2}{dt} = r_2 + \frac{F_{12}}{V_1}x_1 - \frac{F_{21} + F_e}{V_2}x_2. \quad (9)$$

In matrix notation $\frac{d\mathbf{x}}{dt} = A\mathbf{x} + \mathbf{r}$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -k_1 & k_{21} \\ k_{12} & -k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}, \quad (10)$$

where $k_1 = \frac{F_{12}}{V_1}$, $k_{21} = \frac{F_{21}}{V_2}$, $k_{12} = \frac{F_{12}}{V_1}$, and $k_2 = \frac{F_{21} + F_e}{V_2} = \frac{F_{12}}{V_2}$.

Worked Example — Two-Lake Numerical Problem with Complete Solution**Problem Statement:**

Two interconnected lakes have volumes $V_1 = 100 \times 10^6 \text{ m}^3$ and $V_2 = 200 \times 10^6 \text{ m}^3$. Water transfer rates are:

- Fresh water input to Lake 1: $F_0 = 10 \times 10^6 \text{ m}^3/\text{year}$.
- Inter-lake flow from Lake 1 to Lake 2: $F_{12} = 30 \times 10^6 \text{ m}^3/\text{year}$.
- Return pump from Lake 2 to Lake 1: $F_{21} = 20 \times 10^6 \text{ m}^3/\text{year}$.
- Net discharge from Lake 2: $F_e = 10 \times 10^6 \text{ m}^3/\text{year}$.

At $t = 0$, an industrial spill introduces initial pollutant masses $x_1(0) = 500 \text{ kg}$ into Lake 1 and $x_2(0) = 0$ into Lake 2. Further pollutant input is halted ($r_1 = r_2 = 0$). Find:

- The analytical equations for $x_1(t)$ and $x_2(t)$.
- The time $t_{\max}$ when the pollutant in Lake 2 reaches its peak value, and the maximum mass $x_2(t_{\max})$.

Step 1: Compute Rate Constants and Form Matrix A

Using the volumes and flow rates (in units of 10^6 m^3):

$$k_1 = \frac{F_{12}}{V_1} = \frac{30}{100} = 0.3 \text{ yr}^{-1}, \quad k_{21} = \frac{F_{21}}{V_2} = \frac{20}{200} = 0.1 \text{ yr}^{-1},$$

$$k_{12} = \frac{F_{12}}{V_1} = 0.3 \text{ yr}^{-1}, \quad k_2 = \frac{F_{12}}{V_2} = \frac{30}{200} = 0.15 \text{ yr}^{-1}.$$

Thus the homogeneous system is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -0.3 & 0.1 \\ 0.3 & -0.15 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Step 2: Find the Eigenvalues and Eigenvectors

The characteristic equation is $\det(A - \lambda I) = 0$:

$$(-0.3 - \lambda)(-0.15 - \lambda) - (0.1)(0.3) = 0$$

$$\lambda^2 + 0.45\lambda + 0.045 - 0.03 = \lambda^2 + 0.45\lambda + 0.015 = 0.$$

Factoring / Quadratic formula:

$$\lambda = \frac{-0.45 \pm \sqrt{0.45^2 - 4(1)(0.015)}}{2} = \frac{-0.45 \pm \sqrt{0.2025 - 0.0600}}{2} = \frac{-0.45 \pm \sqrt{0.1425}}{2}.$$

Since $\sqrt{0.1425} \approx 0.37749$:

$$\lambda_1 = \frac{-0.45 + 0.37749}{2} \approx -0.03625 \text{ yr}^{-1}, \quad \lambda_2 = \frac{-0.45 - 0.37749}{2} \approx -0.41375 \text{ yr}^{-1}.$$

For eigenvector $\mathbf{v}_k = \begin{bmatrix} u_k \\ w_k \end{bmatrix}$, from the first row $(-0.3 - \lambda_k)u_k + 0.1w_k = 0 \implies w_k = \frac{0.3 + \lambda_k}{0.1}u_k$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ \frac{0.3 - 0.03625}{0.1} \end{bmatrix} = \begin{bmatrix} 1 \\ 2.6375 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ \frac{0.3 - 0.41375}{0.1} \end{bmatrix} = \begin{bmatrix} 1 \\ -1.1375 \end{bmatrix}.$$

Step 3: Apply Initial Conditions

The general solution is:

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + c_2 \mathbf{v}_2 e^{\lambda_2 t}.$$

At $t = 0$, $\mathbf{x}(0) = \begin{bmatrix} 500 \\ 0 \end{bmatrix}$:

$$c_1 + c_2 = 500, \quad 2.6375 c_1 - 1.1375 c_2 = 0.$$

From the second equation, $c_2 = \frac{2.6375}{1.1375} c_1 \approx 2.3187 c_1$. Substituting into the sum:

$$3.3187 c_1 = 500 \implies c_1 \approx 150.66 \text{ kg}, \quad c_2 = 500 - 150.66 = 349.34 \text{ kg}.$$

Therefore, the pollutant profiles are:

$$\begin{aligned} \mathbf{x}_1(t) &= 150.66 e^{-0.03625t} + 349.34 e^{-0.41375t} \quad [\text{kg}], \\ \mathbf{x}_2(t) &= 397.37 (e^{-0.03625t} - e^{-0.41375t}) \quad [\text{kg}]. \end{aligned}$$

Step 4: Peak Pollutant in Lake 2

To find the maximum in Lake 2, set $\frac{dx_2}{dt} = 0$:

$$-0.03625 e^{-0.03625t} - (-0.41375) e^{-0.41375t} = 0 \implies e^{(0.41375-0.03625)t} = \frac{0.41375}{0.03625} \approx 11.4138$$

$$0.3775t = \ln(11.4138) \approx 2.4348 \implies t_{\max} \approx \mathbf{6.45} \text{ years.}$$

The peak mass of pollutant in Lake 2 is:

$$x_2(6.45) = 397.37 (e^{-0.03625(6.45)} - e^{-0.41375(6.45)}) = 397.37(0.7915 - 0.0693) \approx \mathbf{287.0} \text{ kg.}$$

4. Case II: The Three-Lake Cascade (Great Lakes System)

A prominent ecological challenge is the cleanup of unilateral cascades, such as Lake Huron (L_1), Lake Erie (L_2), and Lake Ontario (L_3).

Derivation & Formulation — Lower-Triangular Cascade Model

When water flows sequentially without backflow ($L_1 \rightarrow L_2 \rightarrow L_3$), the transfer matrix is **lower triangular**:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -k_1 & 0 & 0 \\ k_1 & -k_2 & 0 \\ 0 & k_2 & -k_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}, \quad (11)$$

where $k_i = \frac{F}{V_i}$. The eigenvalues are precisely the diagonal entries:

$$\lambda_1 = -k_1, \quad \lambda_2 = -k_2, \quad \lambda_3 = -k_3. \quad (12)$$

When all k_i are distinct, this can be solved sequentially via integrating factors or Laplace transforms.

Key Governing Equation**Step-by-Step Explicit Cascade Solutions (Zero Input $r_i = 0$):**

$$x_1(t) = x_1(0) e^{-k_1 t}, \quad (13)$$

$$x_2(t) = \frac{k_1 x_1(0)}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t}) + x_2(0) e^{-k_2 t}, \quad (14)$$

$$x_3(t) = k_1 k_2 x_1(0) \left[\frac{e^{-k_1 t}}{(k_2 - k_1)(k_3 - k_1)} + \frac{e^{-k_2 t}}{(k_1 - k_2)(k_3 - k_2)} + \frac{e^{-k_3 t}}{(k_1 - k_3)(k_2 - k_3)} \right] \\ + \frac{k_2 x_2(0)}{k_3 - k_2} (e^{-k_2 t} - e^{-k_3 t}) + x_3(0) e^{-k_3 t}. \quad (15)$$

Mathematical Law / Theorem — Half-Life & Cleanup Time Analysis

If Lake i is isolated and the external source is cut off, its cleanup dynamics follow exponential decay:

$$x_i(t) = x_i(0) e^{-k_i t}. \quad (16)$$

- **Half-Life ($t_{1/2}$):** Time required to purge 50% of pollutant:

$$t_{1/2} = \frac{\ln 2}{k_i} = \tau_i \ln 2 \approx 0.693 \tau_i.$$

- **95% Purge Time ($t_{0.05}$):** Time required to remove 95% of pollutant:

$$t_{0.05} = \frac{\ln(1/0.05)}{k_i} = \frac{\ln 20}{k_i} \approx 2.996 \tau_i \approx 3 \tau_i.$$

- **99% Purge Time ($t_{0.01}$):**

$$t_{0.01} = \frac{\ln(100)}{k_i} \approx 4.605 \tau_i.$$

5. Steady-State Equilibrium & Abatement Strategies

When ongoing industrial operations continuously release pollutants at constant rates $\mathbf{r} = [r_1, r_2, \dots, r_n]^T$, the system eventually settles into a time-independent **equilibrium state** $\mathbf{x}^*$.

Definition — Equilibrium Pollution Vector

Setting $\frac{d\mathbf{x}}{dt} = \mathbf{0}$ yields the linear algebraic system:

$$A\mathbf{x}^* + \mathbf{r} = \mathbf{0} \implies \mathbf{x}^* = -A^{-1}\mathbf{r}. \quad (17)$$

The steady-state concentration in each lake is:

$$c_i^* = \frac{x_i^*}{V_i}. \quad (18)$$

Worked Example — Pollution Abatement Strategy Calculation**Scenario:**

Consider a 2-lake serial system where $F = 5 \text{ km}^3/\text{yr}$, $V_1 = 25 \text{ km}^3$, and $V_2 = 50 \text{ km}^3$. A chemical plant currently injects $r_1 = 100 \text{ tons/yr}$ into Lake 1, while Lake 2 receives direct agricultural runoff $r_2 = 25 \text{ tons/yr}$.

- Compute current equilibrium concentrations c_1^* and c_2^* .
- Environmental regulators require c_2^* to not exceed 1.0 ton/km^3 . Determine the necessary percentage reduction in factory emissions r_1 .

Solution:

Rate constants:

$$k_1 = \frac{F}{V_1} = \frac{5}{25} = 0.2 \text{ yr}^{-1}, \quad k_2 = \frac{F}{V_2} = \frac{5}{50} = 0.1 \text{ yr}^{-1}.$$

The steady-state equations $\dot{x}_1 = 0$ and $\dot{x}_2 = 0$:

$$r_1 - k_1 x_1^* = 0 \implies x_1^* = \frac{r_1}{k_1} = \frac{100}{0.2} = 500 \text{ tons}.$$

$$k_1 x_1^* + r_2 - k_2 x_2^* = 0 \implies x_2^* = \frac{k_1 x_1^* + r_2}{k_2} = \frac{100 + 25}{0.1} = \frac{125}{0.1} = 1250 \text{ tons}.$$

Equilibrium concentrations:

$$c_1^* = \frac{x_1^*}{V_1} = \frac{500}{25} = \mathbf{20.0 \text{ tons/km}^3},$$

$$c_2^* = \frac{x_2^*}{V_2} = \frac{1250}{50} = \mathbf{25.0 \text{ tons/km}^3}.$$

Abatement Requirement:

To achieve target concentration $c_2^* \leq 1.0 \text{ ton/km}^3$, the maximum allowed mass is:

$$x_2^* = c_2^* \times V_2 = 1.0 \times 50 = 50 \text{ tons}.$$

From the steady-state equation for Lake 2:

$$r_1 + r_2 = k_2 x_2^* \implies r_1 + 25 = 0.1(50) = 5 \text{ tons/yr.}$$

Notice that even if $r_1 = 0$, the agricultural runoff alone produces $k_2 x_2^* \geq 25 \implies x_2^* \geq 250 \text{ tons}$ ($c_2^* \geq 5.0 \text{ ton/km}^3$).

Conclusion: Regulating the factory alone cannot satisfy the requirement; agricultural runoff r_2 must also be reduced below 5 tons/yr for compliance!

6. Summary of Formulae & Diagnostic Reference

Concept	Mathematical Formulation	Physical Significance
Concentration	$c_i(t) = \frac{x_i(t)}{V_i}$	Mass per unit volume under uniform mixing
Turnover Rate	$k_i = \frac{F_{\text{out},i}}{V_i} = \frac{1}{\tau_i}$	Fraction of volume replaced per unit time
State-Space Matrix	$\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{r}, \quad A_{ii} = -k_i$	Coupling matrix governing system rates
Eigenvalues	$\det(A - \lambda I) = 0 \implies \text{Re}(\lambda_k) < 0$	Exponential relaxation rates (stable)
Equilibrium	$\mathbf{x}^* = -A^{-1}\mathbf{r}$	Long-term stationary pollution level
Half-life	$t_{1/2} = \tau_i \ln 2 \approx 0.693 \tau_i$	Time for 50% single-lake washout
95% Cleanup Time	$t_{0.05} = \tau_i \ln 20 \approx 3.0 \tau_i$	Practical baseline for water restoration

Note for BS Mathematics Students:

The System of Lake Pollution directly integrates linear algebra (matrix diagonalization, Jordan canonical forms, eigenvalues) with systems of ordinary differential equations (integrating factors, fundamental matrix, phase-plane stability). When preparing for university examinations, ensure both the derivation of the mass balance differential equations and the matrix exponential solution technique are thoroughly practiced.